

Contributions of Common, Rare, and Somatic Genetic Variants to Incidence of Atrial Fibrillation

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 Supplemental content

IMPORTANCE Atrial fibrillation (AF) has a complex genetic architecture involving common, rare, and somatic variants. The association between these components requires further investigation.

OBJECTIVE To examine the individual and combined contributions of polygenic, monogenic, and somatic genetic variants to AF incidence, and develop an integrated genomic model (IGM-AF) for improved risk prediction.

DESIGN, SETTING, AND PARTICIPANTS This cohort study used whole-genome sequence data from participants of the UK Biobank, with follow-up for AF events through hospital records, death registries, and self-report. The UK Biobank recruited participants aged 40 to 69 years in the UK between 2006 and 2010. Study data were analyzed from August 2022 to November 2024.

EXPOSURES IGM-AF comprising an AF polygenic risk score (PRS), a composite rare variant gene set (AFgeneset), and somatic variants associated with clonal hematopoiesis of indeterminate potential (CHIP). Clinical AF risk was estimated using the Cohorts for Heart and Aging Research in Genomic Epidemiology AF (CHARGE-AF) score.

MAIN OUTCOMES AND MEASURES The primary outcome was hazard ratios (HRs) for 5-year incident AF attributable to PRS, AFgeneset, CHIP, and their interactions. The predictive performance of IGM-AF and its components was quantified using HRs, C statistics, and reclassification indices.

RESULTS A total of 416 085 individuals (mean [SD] age, 56.6 [8.0] years; 224 642 female [54.0%]) with 30 797 AF cases were included. The PRS (HR per 1 SD, 1.65; 95% CI, 1.63-1.67; $P < 1 \times 10^{-8}$), AFgeneset (HR, 1.63; 95% CI, 1.52-1.75; $P = 1.46 \times 10^{-42}$), and CHIP (HR, 1.26; 95% CI, 1.15-1.38; $P = 1.41 \times 10^{-6}$) were associated with incident AF. The 5-year cumulative incidence of AF was at least 2-fold among individuals having all 3 genetic drivers (common, rare, and somatic drivers) compared with those with only 1 driver. Integration of IGM-AF with a clinical risk model (CHARGE-AF) showed higher predictive performance (C statistic, 0.80; 95% CI, 0.80-0.80) compared with IGM-AF and CHARGE-AF alone. The classification of the at-risk population for AF was improved when IGM-AF was added to CHARGE-AF (net reclassification index, 0.08; 95% CI, 0.07-0.09).

CONCLUSIONS AND RELEVANCE Results of this cohort study demonstrated the complementary value of common, rare, and somatic variants in shaping genomic AF risk. Leveraging comprehensive genetic information may enhance screening and preventive interventions for AF.

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Atrial fibrillation (AF) is one of the most prevalent cardiac arrhythmias, leading to increased risks of stroke, heart failure, and premature death.¹ In recent decades, significant advances have been made in understanding the genetic contributions to AF, including common and rare germline variants.² Genome-wide association studies (GWAS) have identified hundreds of loci associated with AF,³ highlighting common variants with modest effect sizes. Rare germline variants with larger effects, particularly those causing loss of function (LoF), have also been implicated, precipitating early-onset AF risk in certain subpopulations.^{4,5} Despite this growing knowledge of germline variation predisposing to AF, deciphering how polygenic and monogenic elements individually and collectively contribute to AF remains to be understood.

Somatic gene variants—particularly clonal hematopoiesis of indeterminate potential (CHIP)—add another layer of complexity to the genetic landscape of AF.^{6,7} Clonal hematopoiesis in *TET2* and *ASXL1* has been associated with increased risk of AF.⁶ Lin et al⁷ further revealed a significant influence of *TET2*-derived CHIP on AF risk that is likely driven by *Nlrp3* inflammasome activation. Germline components (common and rare variants) are inherited, whereas somatic gene variants are acquired over the course of aging.⁸ Nonetheless, they are not independent, as growing evidence has revealed that certain germline variants, particularly in DNA repair genes, can strongly predispose to somatic variants.^{8,9} However, potential relations between somatic variants and germline variation with respect to AF risk have not been adequately explored.

Leveraging whole-genome sequence data of over 400 000 individuals from the UK Biobank, we pursued an integrated exploration of all known genetic drivers of AF—common, rare, and somatic variants. By unifying germline and somatic information into a comprehensive framework, we aimed to provide a more complete understanding of the genetic contributions to AF and illustrate how these components may independently and synergistically modulate AF susceptibility.

Methods

Dataset Availability and Quality Control

The use of UK Biobank data was approved under application ID 89885, and all participants provided written informed consent to participate. This study was conducted with approval from the Ethics Committee of the Beijing Institute of Genomics, Chinese Academy of Sciences, and the China National Center for Bioinformation. Quality control filters were applied to the following variants and samples: variant call rate (<95%), sample call rate (<95%), Hardy-Weinberg equilibrium test ($P < 1 \times 10^{-6}$), outliers for heterozygosity, and discordance between genetically inferred sex and self-reported or ambiguous sex ($0.5 < F$ statistic < 0.8)¹⁰; 1 of the samples whose pairwise kinship value was greater than or equal to 0.0884 (defined by second degree of relationship or closer) was removed.¹¹ Participant race and ethnicity were directly derived from the UK Biobank and included Asian, Black, Chinese, multiracial/

Key Points

Question What are the contributions of common, rare, and somatic variations, and their association with the incidence of atrial fibrillation (AF)?

Findings This cohort study including 416 085 individuals demonstrated the individual and complementary roles of common, rare, and somatic variations for AF onset. The rare variants, polygenic score, and clonal hematopoiesis of indeterminate potential enhanced the prediction of incident AF when added to clinical factors.

Meaning Results suggest that leveraging comprehensive genetic information may aid in AF prevention and risk stratification.

ethnic, White, and other, for which no further subdivisions were specified. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

Common, Rare, and Somatic Variants

To capture the proportion of AF explained by common variants, a polygenic risk score (PRS) was calculated using variant weights derived from our multiethnic GWAS for AF,¹² implemented by PLINK.¹³ We used UK Biobank-excluded PRS weight file to calculate AF PRS for UK Biobank participants, minimizing the risk of overfitting.¹² To control for potential confounding effects, we regressed PRS on recruitment age, sex, and the first 10 principal components (PCs) of genetic ancestry. The resulting residuals were standardized to have 0 mean and unit SD (eFigure 2 in Supplement 1), and these scaled values were used in subsequent analyses as previously performed.¹⁴

To determine the carrier status of rare variants contributing to AF incidence, rare variants annotated as LoF, which are typically predicted to severely disrupt gene function, were captured by a multistep selection strategy. We first identified genes previously reported to be associated with AF. For genes with available rare LoF variant lists from prior studies,² those variants were directly included. For genes without predefined variant sets,⁵ high-confidence LoF variants were annotated by the LoF Transcript Effect Estimator plugin within the Ensembl Variant Effect Predictor.¹⁵ Next, LoF variants with a minor allele frequency of less than 0.01 located within the same gene were collapsed for burden test by Regenie,¹⁶ to validate their association with AF. Both prevalent and incident AF cases were included to increase statistical power through a larger sample size. Genes that passed this validation were included in the final AFgeneset. At the individual level, a binary carrier status was assigned: participants carrying at least 1 qualifying LoF variant in any of the AFgeneset genes were coded as *carrier*, and those without were coded as *noncarrier*. The full list of associated rare LoF variants is presented in the eMethods in Supplement 1.

The somatic CHIP variants were curated after a stepwise method that combined filtering based on sequencing metrics, variant annotation, and population-based associations.¹⁷ CHIP is most commonly driven by genes involved in epigenetic and transcriptional regulation (such as *TET2*, *ASXL1*, *JAK2*),

messenger RNA splicing (*SF3B1* and *SRSF2*), or DNA damage repair (*TP53* and *PPM1D*).¹⁷ We performed gene-level validation of these 7 genes by Cox proportional hazards models to assess associations. Carrier status was defined as the presence of at least 1 associated CHIP variant.

Statistical Analysis

Identifying Population Prevalence and Estimating Effect Sizes

To measure the population prevalence of genetic drivers, we calculated characteristics of PRS quintiles, carrier status of AFgeneset, and carrier status of CHIP in the entire baseline cohort, including both prevalent AF and incident AF cases, while excluding prevalent AF cases in all subsequent prediction analyses. AF was defined based on self-report at enrollment, hospitalization records, or death registry records. The corresponding codes are described in eTable 1 in Supplement 1.

To estimate the effect sizes of individual genetic drivers for incident AF, a Cox proportional hazards model was used with individual genetic drivers as predictors, incident AF as the outcome, and the age, sex, first 10 PCs of genetic ancestry as covariates. The interactions between pairwise genetic drivers were evaluated using both multiplicative and additive interaction. Multiplicative interactions were assessed by including an interaction term within the Cox proportional hazards model. Additive interactions were quantified using the relative excess risk due to interaction (RERI), where RERI >0 indicates significant additive interaction effects.¹⁸

Developing the Integrated Genomic Model

We systematically combined the 3 genetic drivers (PRS, AFgeneset, and CHIP) into an integrated genomic model for AF (IGM-AF). The combination weights were estimated by a joint regression model. Specifically, PRS, AFgeneset, and CHIP were fitted to a Cox proportional hazards model, setting incident AF as an outcome, and the corresponding Cox coefficients were used as weights to calculate a weighted summation. The weighted summation was then regressed on recruitment age, sex, and the first 10 PCs, and the resulting residuals were scaled to mean 0 and an SD of 1. These scaled values were used in subsequent analyses.

Interplay Between Genetic Drivers and Clinical Risk Score

We then investigated the interaction between genetic risk drivers and a guideline-based clinical risk score for incident AF prediction. Cohorts for Heart and Aging Research in Genomic Epidemiology AF (CHARGE-AF) was a 5-year predictive model including the AF relevant clinical variables (eTable 2 in Supplement 1). Aligning with the 5-year prediction of CHARGE-AF, we defined incident AF cases strictly here as participants who developed AF within the first 5 years of follow-up. Participants developing AF beyond this 5-year window were right censored at 5 years and were treated as the control group in our primary analysis, ensuring consistent window period definition across studies.

We evaluated the discrimination of different models by comparing the C statistics, which was calculated by pROC package in R, version 4.2.2 (R Project for Statistical Computing),

Table. Participant Characteristics for UK Biobank

Characteristic	No. (%)
No.	416 085
Age, mean (SD), y	56.59 (8.03)
Sex, No. (%)	
Female	224 642 (54.0)
Male	191 443 (46.0)
Follow-up duration, median (IQR), y	13.2 (12.4–14.0)
Race and ethnicity, No. (%)	
Asian	8011 (1.9)
Black	6516 (1.6)
Chinese	1360 (0.3)
Multiracial/ethnic	2314 (0.6)
White	392 287 (94.3)
Other ^a	5597 (1.3)
Atrial fibrillation, No. (%)	30 797 (7.4)
Prevalent atrial fibrillation	8558 (2.1)
Incident atrial fibrillation	22 239 (5.3)
Incident atrial fibrillation within the 1st 5 y of follow-up	7450 (1.8)
Smoking, No. (%)	43 130 (10.4)
BMI, mean (SD) ^b	27.41 (4.75)
Measured SBP, mean (SD), mm Hg	137.92 (18.60)
Measured DBP, mean (SD), mm Hg	82.29 (10.14)
Antihypertensive medication, No. (%)	47 204 (11.3)
Diabetes, No. (%)	9607 (2.3)
History of myocardial infarction, No. (%)	9968 (2.4)
History of heart failure, No. (%)	2165 (0.5)

Abbreviations: BMI, body mass index; DBP, diastolic blood pressure; SBP, systolic blood pressure.

^a Other: self-defined ancestry not included in Asian, Black, Chinese, multiracial/ethnic, or White.

^b Calculated as weight in kilograms divided by height in meters squared.

and the 95% CIs of C statistics were computed with stratified bootstrap. The statistically significant improvement between different models was defined as a bootstrap-estimated (500 iterations) difference in C statistics (delta C statistic) with a 95% CI that did not cover 0.¹⁹

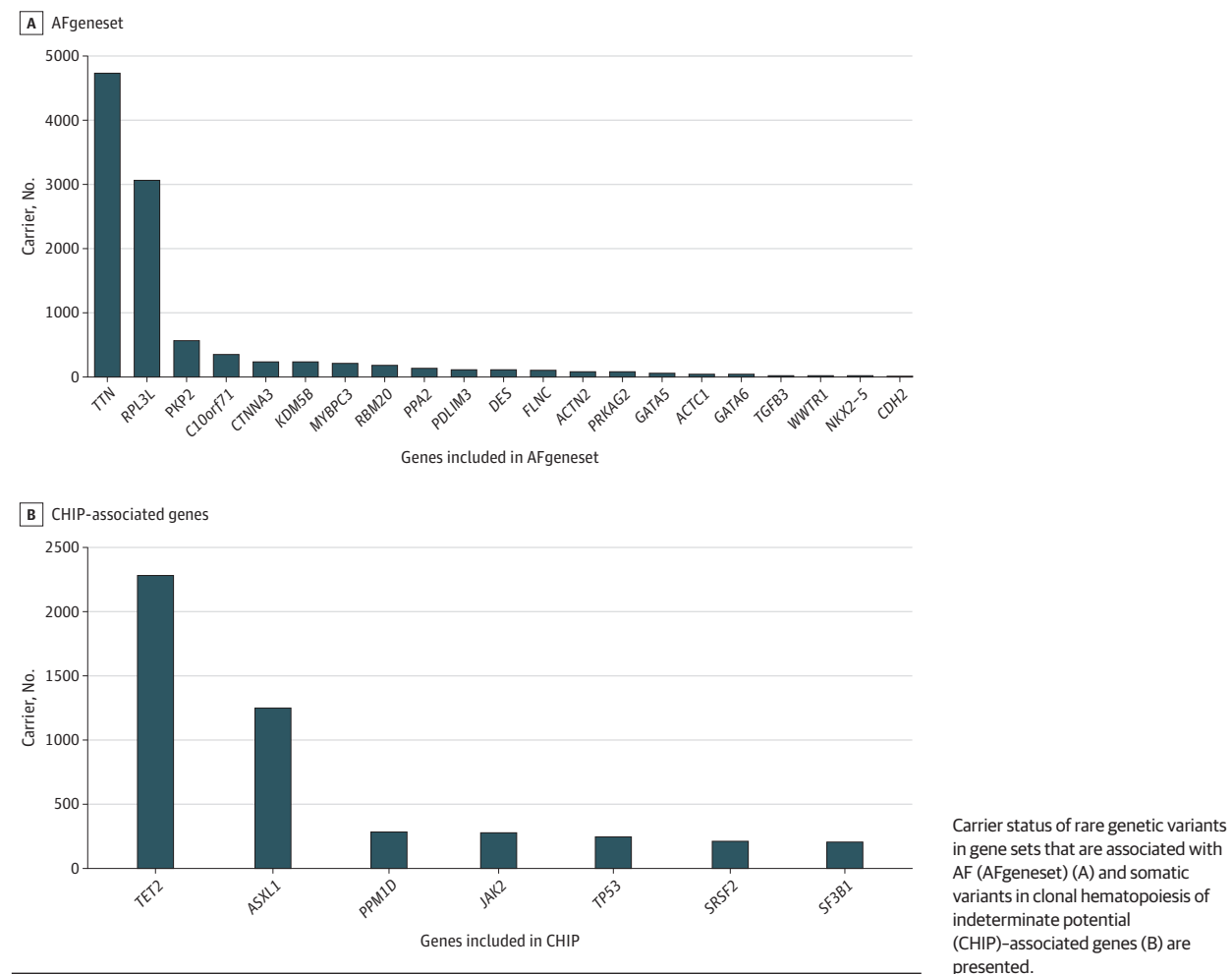
We then assessed the potential prediction power improvement with addition of IGM-AF to CHARGE-AF by estimating net reclassification index (NRI), with a threshold of 5% for categorical reclassification and threshold 0 for continuous reclassification, as performed in previous studies.²⁰ All statistical analyses were performed from August 2022 to November 2024 using R, version 4.2.2 (R Project for Statistical Computing). All *P* values were 2-sided, and *P* < .05 was considered statistically significant. A full method description can be found in eMethods in Supplement 1.

Results

Baseline Characteristics

A total of 416 085 individuals (mean [SD] age, 56.6 [8.0] years; 224 642 female [54.0%]; 191 443 male [46.0%]) with 30 797 AF

Figure 1. Carrier Status of Rare and Somatic Variants Associated With Atrial Fibrillation (AF)



cases were included in the analysis after quality control (Table and eFigure 1 in Supplement 1). Races and ethnicities of the individuals were as follows: 8011 Asian (1.9%), 6516 Black (1.6%), 1360 Chinese (0.3%), 2314 multiracial/ethnic (0.6%), 392 287 White (94.3%), and 5597 other (1.3%). There were 8558 individuals (2.1%) with AF before enrollment in the UK Biobank, an additional 7450 participants (1.8%) who developed AF over the first 5 years of follow-up, and 22 239 participants (5.3%) who developed AF over a total median (IQR) follow-up of 13.2 (12.4-14.0) years (Table).

Characteristics of Genetic Drivers

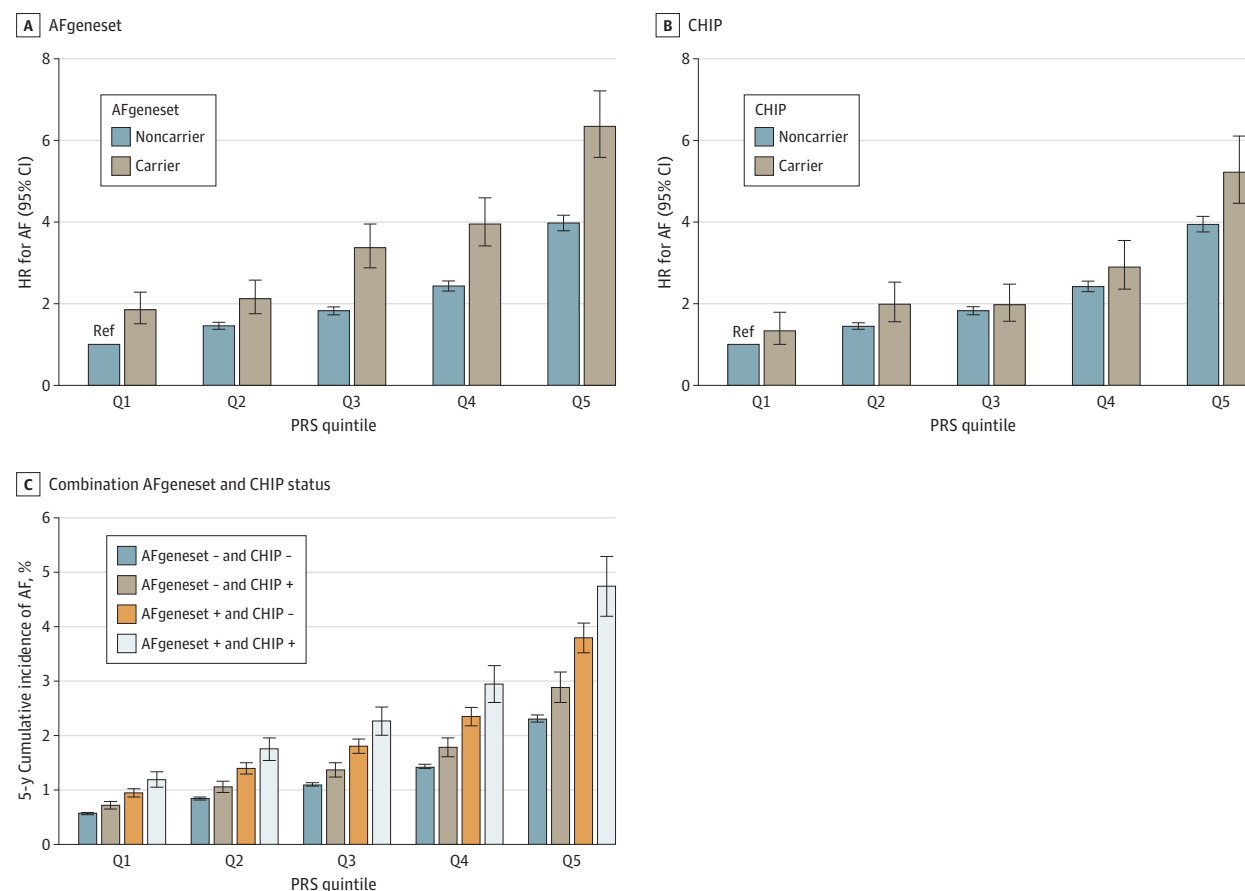
As for common variants, the proportion of individuals with AF increased progressively across ascending PRS quintiles (eTable 3 in Supplement 1). As for rare variants, the 21 genes included in AFgeneset (*TTN*, *KDM5B*, *C10orf71*, *CTNNA3*, *PKP2*, *RPL3L*, *MYBPC3*, *DES*, *RBM20*, *FLNC*, *PRKAG2*, *ACTC1*, *ACTN2*, *GATA6*, *GATA5*, *NKX2-5*, *TGFB3*, *CDH2*, *PPA2*, *WWTR1*, *PDLIM3*) were all significantly associated with AF (eTable 4 in Supplement 1), and the number of carriers for these genes ranged from 9 for *CDH2* to 4720 for *TTN* (Figure 1A). Rare variants in the AFgeneset were identi-

fied in 1186 individuals (3.9%) in the AF group, compared with 8610 individuals (2.2%) in the non-AF group (eTables 5 and 6 in Supplement 1). As for somatic variants of the 7 genes included in CHIP (*TET2*, *ASXL1*, *JAK2*, *PPM1D*, *TP53*, *SF3B1*, *SRSF2*), 6 were significantly associated with AF (all except *ASXL1*) (eTable 7 in Supplement 1), and the number of carriers for these genes ranged from 186 to 2283 (Figure 1B). Somatic CHIP variants were detected in 644 individuals (2.1%) in the AF group compared with 3919 individuals (1.0%) in the non-AF group (eTables 5 and 8 in Supplement 1).

Association of Genetic Drivers With AF Incidence

Each genetic driver was significantly associated with incident AF (eFigure 3 in Supplement 1): PRS (hazard ratio [HR] per 1 SD, 1.65; 95% CI, 1.63-1.67; $P < 1 \times 10^{-8}$), AFgeneset (HR, 1.63; 95% CI, 1.52-1.75; $P = 1.46 \times 10^{-42}$), and CHIP (HR, 1.26; 95% CI, 1.15-1.38; $P = 1.41 \times 10^{-6}$). We also explored the association of these genetic drivers with other important clinical outcomes (eTable 9 in Supplement 1). For heart failure, AF PRS was associated with an increased hazard (HR per 1 SD, 1.15; 95% CI, 1.13-1.17; $P = 2.16 \times 10^{-49}$), as were AFgeneset (HR, 1.63; 95% CI, 1.47-

Figure 2. Effect of Gene Sets Associated With Atrial Fibrillation (AF; AFgeneset) and Clonal Hematopoiesis of Indeterminate Potential (CHIP) With Atrial Fibrillation Stratified by Polygenic Risk Score (PRS)



A, AFgeneset. B, CHIP. Noncarriers in the lowest PRS quintile were set as the reference group. C, 5-Year cumulative incidence of atrial fibrillation based on combinations of carrier status for AFgeneset and CHIP variants, stratified by PRS. The 5-year cumulative incidence of AF ranged from 0.6% in the lowest group to 4.7% in the highest group. The + sign indicates positive carrier status

and the - sign indicates negative carrier status. The 5-year cumulative incidence of AF was estimated by the Cox proportional hazards model with individual genetic driver as predictor, incident AF as outcome, and the age, sex, first 10 principal components of genetic ancestry as covariates. The bar represents 95% CI. HR indicates hazard ratio.

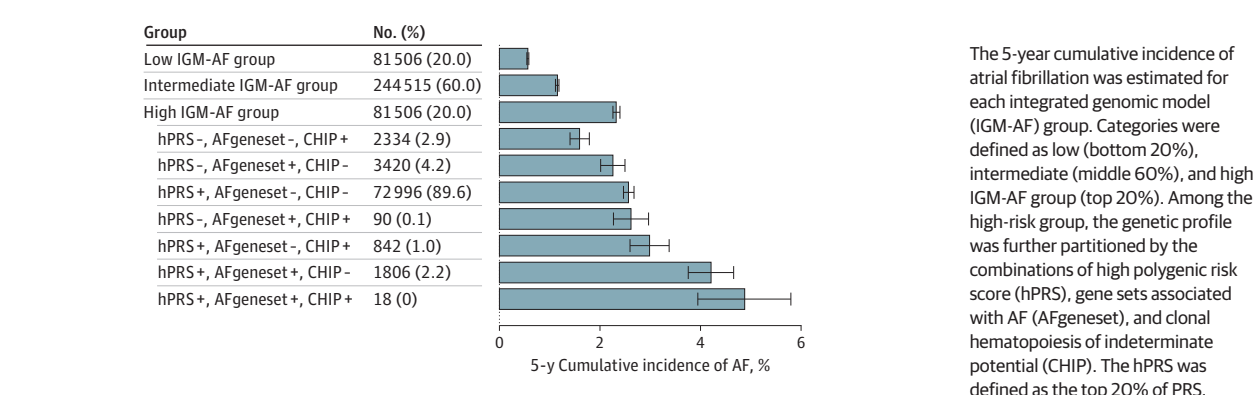
1.79; $P = 2.02 \times 10^{-22}$) and CHIP (HR, 1.60; 95% CI, 1.42-1.80; $P = 1.07 \times 10^{-14}$). For ischemic stroke, PRS was significantly associated (HR per 1 SD, 1.12; 95% CI, 1.09-1.15; $P = 1.26 \times 10^{-18}$). AFgeneset also showed a significant association (HR, 1.36; 95% CI, 1.17-1.57; $P = 4.19 \times 10^{-5}$), whereas the association between CHIP and ischemic stroke did not meet conventional statistical significance. These drivers also showed an association with coronary artery disease, although to a lesser extent compared with the other 2 outcomes.

For the interaction analysis of 3 genetic drivers, no significant interaction was observed between AFgeneset and CHIP on either the additive (RERI, 0.12; 95% CI, -0.79 to 1.04) or multiplicative scales. Significant additive interactions, however, were found between PRS and AFgeneset (RERI, 0.25; 95% CI, 0.19-0.31) and between PRS and CHIP (RERI, 0.10; 95% CI, 0.04-0.15), respectively. These results suggest that the combined effect of a high PRS carrier with AFgeneset or CHIP exceeds the sum of each effect (Figure 2).

Integration of Genetic Drivers to Predict AF Incidence

To have a comprehensive assessment of common, rare, and somatic variants in prediction of incident AF, we ensembled PRS, AFgeneset, and CHIP. For the sex-combined estimation of 5-year cumulative incidence of AF (Figure 2C), individuals in the lowest PRS quintile without AFgeneset and CHIP variants had a 5-year AF incidence as low as 0.6%, whereas those in the highest PRS quintile carrying AFgeneset and CHIP variants had a 5-year incidence as high as 4.7%, the risk difference around 7.8 fold. For the sex-stratified analysis, a similar trend was observed, but males had a 2-fold higher AF incidence compared with females (eFigure 4 in Supplement 1). A similar trend was observed for cumulative incidence of AF for different combinations (eFigures 5 and 6 in Supplement 1). The AF incidence increases with the number of genetic drivers. For example, among the high IGM-AF group, individuals carrying all 3 types of genetic drivers had the highest 5-year cumulative incidence, followed by those carrying 2 types

Figure 3. Diverse Genetic Risk Profiles Contributing to Atrial Fibrillation (AF)



of genetic drivers, and then those carrying only 1 type of genetic driver, reflecting an additive and diverse profile of genetic drivers for predisposing AF (Figure 3 and eTable 10 in Supplement 1).

Genetic and Clinical Risk for AF

For the interaction between genetic and clinical risk, the 3 genetic drivers effectively stratified AF incidence in both the low (<5%) and high CHARGE-AF (>5%) strata (Figure 4A-C). In the high CHARGE-AF stratum, the estimated 5-year incidence of AF were 3.22% (95% CI, 3.01%-3.42%), 6.68% (95% CI, 6.39%-6.96%), and 13.58% (95% CI, 12.94%-14.22%) for low, intermediate, and high PRS groups, respectively. In contrast, the numbers were 0.61% (95% CI, 0.58%-0.65%), 1.29% (95% CI, 1.25%-1.33%), and 2.70% (95% CI, 2.60%-2.80%), respectively, in the low CHARGE-AF stratum (Figure 4A and eTable 11 in Supplement 1), implicating 5-fold lower risk compared with the high CHARGE-AF group. For noncarriers and carriers of AFgeneset, the estimated 5-year incidence of AF was 7.22% (95% CI, 6.92%-7.52%) and 12.44% (95% CI, 11.05%-13.81%), respectively, in the high CHARGE-AF stratum, and 1.41% (95% CI, 1.37%-1.45%) and 2.49% (95% CI, 2.21%-2.77%) in the low CHARGE-AF stratum (Figure 4B and eTable 12 in Supplement 1). For noncarriers and carriers of CHIP, the estimated 5-year incidence of AF was 7.26% (95% CI, 6.95%-7.57%) and 10.53% (95% CI, 8.90%-12.12%), respectively, in the high CHARGE-AF stratum, and 1.43% (95% CI, 1.39%-1.47%) and 2.10% (95% CI, 1.77%-2.44%), respectively, in the low CHARGE-AF stratum (Figure 4C and eTable 13 in Supplement 1). The IGM-AF, which combined the 3 genetic drivers, showed an even more pronounced gradient in each stratum of the CHARGE-AF (eFigure 7 in Supplement 1). Significant multiplicative ($P = 3.44 \times 10^{-12}$) and additive interactions (RERI, 1.04; 95% CI, 0.96-1.11) were observed between IGM-AF and CHARGE-AF, a much broader gradient for the high CHARGE-AF compared with the low CHARGE-AF group.

By adding the ensemble predictor IGM-AF to CHARGE-AF, the combined model improved the conventional clinical model in AF incidence prediction (Figure 4D). The C statistic for CHARGE-AF was 0.77 (95% CI, 0.76-0.77), IGM-AF was 0.77 (95% CI, 0.76-0.77), and the combination of CHARGE-AF and IGM-AF model was 0.80 (95% CI, 0.80-0.80) (Figure 4D). The

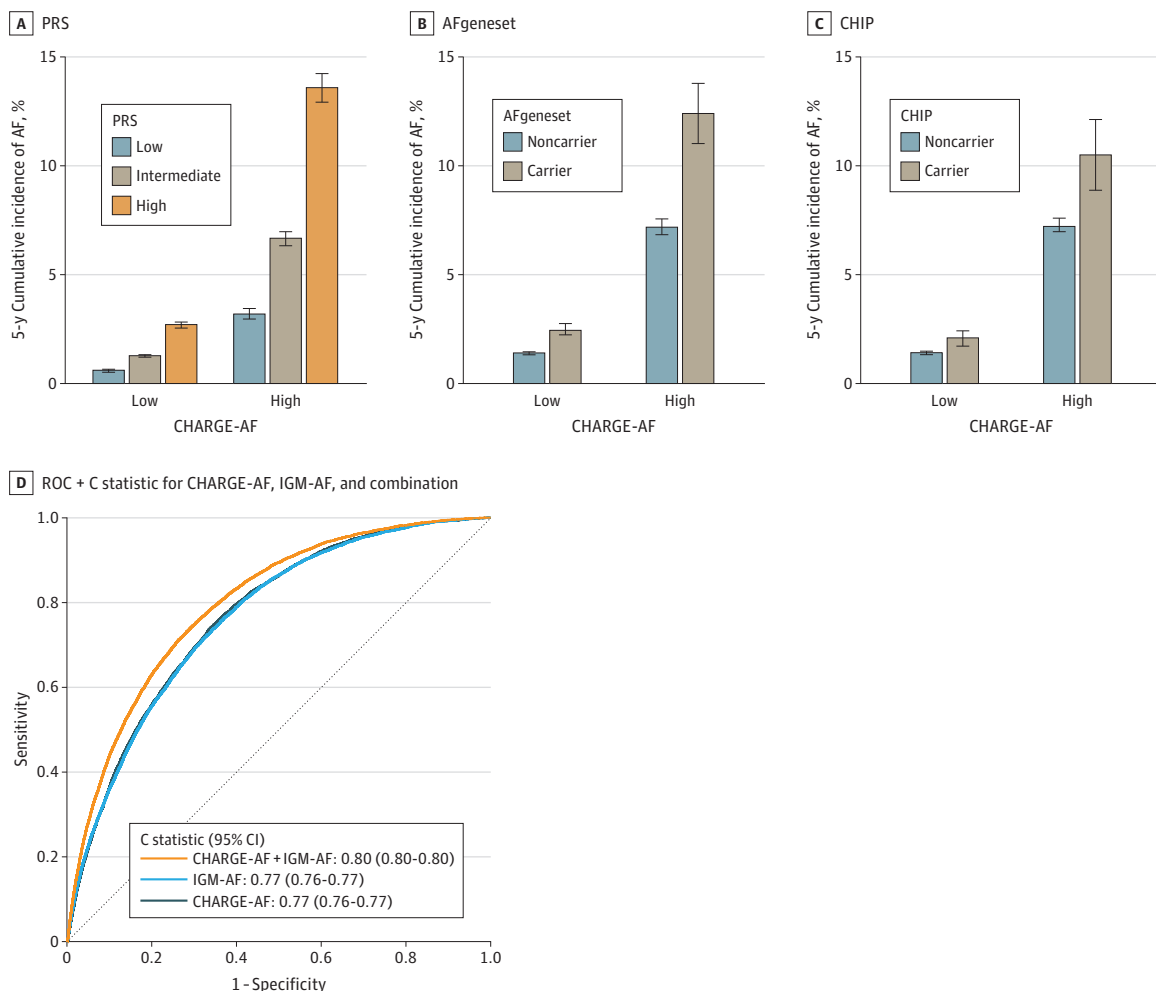
combined model also showed strong reclassification capacity. When adding IGM-AF to CHARGE-AF compared with CHARGE-AF alone, the overall NRI resulted in 0.08 (95% CI, 0.07-0.09) at the 5% threshold model (eTable 14 in Supplement 1). The continuous NRI was 0.43 (95% CI, 0.40-0.45). When comparing the model prediction with the actual event rate, a close match of the model predicted 5-year incidence and actual observed 5-year incidence of AF shows the models were well calibrated (eFigure 8 and eTable 14 in Supplement 1).

Discussion

By exploring common, rare, and somatic variants, we uncovered their individual and collective roles in shaping AF risk. Harnessing the complete range of genetic data provided deeper insights into AF and captured high-risk individuals who might otherwise be overlooked when looking at genetic drivers in isolation. Moreover, to demonstrate how various genetic factors could be integrated, we derived IGM-AF, a unified score. Addition of the IGM-AF to the CHARGE-AF substantially improved classification of high-risk individuals, underscoring the potential clinical value of incorporating comprehensive genomic information with clinical risk profiles.

The synergistic interplay between common and rare germline variants in AF may imply an effect modification, where polygenic risk influences the effect of monogenic variants. We previously demonstrated that polygenic risk likely alters the penetrance of monogenic variants, thereby modifying the overall risk profile for common diseases.²¹ With regard to AF specifically, we observed that the polygenic risk significantly modified penetrance of *TTN* variants—carriers of *TTN* LoF variants in the highest tertile of the AF PRS had an AF prevalence of 21.5% compared with 6.7% among the carriers in the lowest AF PRS tertile.²² From a biological perspective, further research is required to understand how the substantial disruptions in a specific molecular pathway introduced by monogenic variants can be exaggerated or counterbalanced by polygenic risk potentially reflecting a broad range of physiological processes.²¹ Given that numerous local molecular pathways are known to converge to influence broader metabolic or functional networks,^{23,24} disruptions in different path-

Figure 4. Interplay of Genetic and Clinical Drivers on Atrial Fibrillation (AF)



5-Year cumulative incidence of AF gradient stratified by polygenic risk score (PRS) (A), gene sets associated with AF (AFgeneset) (B), and clonal hematopoiesis of indeterminate potential (CHIP) (C) within Cohorts for Heart

and Aging Research in Genomic Epidemiology AF (CHARGE-AF) stratum. D, Receiver operating characteristic (ROC) curves and C statistics for CHARGE-AF, integrated genomic model (IGM-AF), and their combination.

ways caused by rare and common variants may collectively impact a higher-level cellular network or disease program.²⁴ Another possible explanation would be the liability threshold model, where disease develops when the combined liability surpasses a certain threshold.²⁵ When polygenic risk is already high, individuals with an AF monogenic load might more easily exceed the threshold needed for AF manifestation. This phenomenon aligned with our observation that AF incidence increases with the number of genetic drivers accumulated (Figure 3).

Polygenic background of AF may also predispose individuals to somatic variants. Although not as pronounced as the interaction between PRS and rare variants, a significant additive interaction was also evident between PRS and CHIP. Our observation aligns with a previous study⁹ suggesting that germline variations contribute to the emergence of CHIP. Chronic infections or sterile inflammatory conditions have shown to increase cellular stress, precipitating CHIP, especially in those with an underlying germline predisposition.^{26,27}

To fully leverage the genetic contributions and interactions among common, rare, and somatic genomic drivers, we developed an integrated genomic model, IGM-AF. This model aimed to translate the complex genetic underpinnings of AF into a clinically actionable framework by providing a singular estimate for AF. It becomes clinically deliverable only when all the genomic information is combined to yield a single estimate, analogous to clinical risk calculators that synthesize multiple clinical risk factors into a single risk score.^{28,29} This model appeared to successfully capture the cumulative effect of common, rare, and somatic genetic factors, demonstrating a dose-dependent increase in 5-year cumulative incidence of AF across incremental genetic risk categories; the high genetic group (top 20%) identified by IGM-AF showed over 2-fold greater 5-year cumulative incidence of AF than the population average group (IGM-AF middle 60%) (Figure 3).

The IGM-AF emphasized the significance of genetic combinations over individual genetic drivers in predicting AF out-

come. The 5-year cumulative incidence of AF was approximately 2-fold higher in individuals having all 3 genetic drivers (common, rare, and somatic drivers) compared with those with only 1 driver, although careful interpretation is warranted due to a small number of people carrying all 3 drivers ($n = 18$). Although the effects of having 1 or all 3 drivers were straightforward, 2-driver combinations revealed more intricate patterns. For example, among individuals with high IGM-AF, those carrying rare and CHIP variants had comparable AF incidence with those with high PRS and CHIP variants; however, individuals with high PRS and rare variants had significantly higher AF incidence compared with other 2-driver combinations (Figure 3). To our knowledge, this was the first study to assess AF risk stratification on the basis of risk spanning 3 major genetic contributors.

The integration of the IGM-AF with a clinical risk score resulted in higher predictive performance than either genetic or clinical score alone, underscoring the complementary role of genetic data in clinical risk assessment. By providing further risk stratification within each clinical risk group, the IGM-AF enhanced the precision of identifying individuals at varying risk levels (Figure 4). Notably, some proportions of low genetic risk individuals classified as high risk by CHARGE-AF had an equivalent or lower 5-year AF risk to high genetic risk individuals in the low CHARGE-AF group (eFigure 7 in Supplement 1). This underscores the dynamic shifts in risk stratification when genetic and clinical factors are considered together, with certain individuals reclassified between high- and low-risk groups. Moreover, the addition of the IGM-AF to the clinical risk model resulted in an NRI of 0.08. This highlighted the benefit of making the most of all available genetic information alongside clinical factors for comprehensive AF risk evaluations.

Limitations

Despite the advances demonstrated in our study, some limitations must be acknowledged. First, although we integrated both common and rare genetic variants into our analysis, the

functional consequences of many rare variants remain to be understood. Although stringent criteria were applied for gene selection, we acknowledge the potential for circularity bias due to the inclusion of certain AF-associated genes initially identified from UK Biobank analyses. To address this limitation, further research is needed to elucidate the precise mechanisms through which these variants contribute to AF risk and progression, future efforts should prioritize validation in ethnically diverse cohorts and mechanistic investigations using experimental models to clarify causal relationships. Second, the use of a single time point DNA sample per individual restricted our ability to explore CHIP dynamics. Although temporal changes in CHIP status could influence predictive performance over extended follow-up, the relatively infrequent acquisition of CHIP in younger to middle-aged individuals minimizes the likelihood of significant impact on our model's utility in this population.³⁰ Last, although IGM-AF demonstrated improved predictive performance in the UK Biobank, validation in more diverse populations is necessary to ensure the generalizability of our findings. We have not sought out external validation of the integrated model, as our intention was to demonstrate the collective contribution of these genetic factors and to enable composite risk assessment, rather than to develop a prediction model for immediate clinical application.

Conclusions

Leveraging the full spectrum of genetic information delivers a deeper understanding of AF development and enhances the accuracy of established clinical assessments. In this cohort study through examining common, rare, and somatic variants at ground level, we demonstrated how these variants contribute uniquely and synergistically to AF. Although a single genomic model integrating different genetic factors demonstrated the complementary and additive nature of genetic risks, the clinical utility of the model is yet uncertain.

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